

TWINKIT · REQUIREMENTS

TwinKit requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/twinkit		
Source	docs/03-requirements.md @ 652cdc0		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Status from the TRL 3 calculations (TWK-CAL-001); R1 found not met at SF9; decisions per TWK-DDR-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

TwinKit requirements

Status is now judged from the TRL 3 calculation note TWK-CAL-001 and the parametric model. Two requirements are not met: R1 (uplink loss 1.02 % against 1 % if all 50 nodes use SF9 at 5 min, a small margin) and R14 (parts \$290 against \$200). R12 (no processor throttling at 40 °C) is at risk, and R13 (setup time) can only be shown by a timed trial. No target was changed at this revision. The design choices adopted in TWK-DDR-001 do not relax or redefine any target; a \$300 budget for R14 is recommended and remains proposed, awaiting Amish.

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 3 (TWK-CAL-001)	Verification (TRL 4 or later)
R1	Receive readings from lab sensor nodes	50 nodes at 1 reading per 5 min each, under 1 % of uplinks lost at SF7 to SF9	Not met at SF9 (marginal): 0.30 % at SF7, 0.56 % at SF8, 1.02 % at SF9 with all nodes on one SF; 0.21 % for an SF7 to SF9 mix; 48 nodes at SF9 meet it [A2, A3, A5]	Bench and field packet counts
R2	Accept common inputs	LoRaWAN (EU868 or US915), MQTT over Ethernet or Wi-Fi, HTTP POST	Met by design	Configuration check
R3	Store history on the gateway	1 year for 50 nodes x 10 fields at 5 min intervals	Met: 4.73 GB per year uncompressed on 56 GB free [B1, B2]; card wear 5.5 to 8.2 card cycles per year, to watch [B4]	Measured database size
R4	Link data to the model	Every sensor channel mapped to a named part (BOM number) of the project's build123d model and to an expected value or calculation, in a validated schema	Met by design (schema fields adopted, TWK-DDR-001)	Schema review on the FieldNode example twin
R5	Compare measured and expected	Residual and pass or flag state shown within 60 s of a reading arriving; flag when outside a stated tolerance for a stated time	Met: 0.9 to 4.7 s [C1]	Bench run
R6	Show the twin in 3D	The project's glTF model, colored by live values, in a current browser on a mid-range laptop or phone	Met by design; frame rate not verifiable at TRL 3	Browser trial
R7	Work offline	All functions run with no internet; optional upstream sync buffers at least 7 days	Met by design; 7-day buffer 91 MB [B3]	Offline run
R8	Low-voltage power	9 to 30 V DC input, 12 V nominal; average draw 8 W or less	Met: 6.41 W average; 20.6 W peak, 2.29 A at 9 V, 3.15 A T fuse [D1 to D3]	Power measurement
R9	Ride through outages	At least 30 min on backup and a clean shutdown before the pack is empty	Met: 2.40 h new, 1.63 h aged pack at 0 °C [E1]	Timed outage
R10	Compact mounting	Fits a TS35 DIN rail; gateway, converter and backup within 20 modules (350 mm) of rail	Met: 294 mm, 16.8 modules [H1]	Fit check
R11	Secure by default	Unique credentials set at first boot, TLS on the dashboard, no ports open beyond the local network by default	Met by design	Configuration audit
R12	Operate in a lab or equipment room	Ambient 0 to 40 °C without CPU throttling	At risk: processor 69.1 °C at light load, 96.9 °C at sustained full load, vented, against an assumed 85 °C throttle point [F2]	Thermal test at 40 °C

ID	Requirement	Target	Status at TRL 3 (TWK-CAL-001)	Verification (TRL 4 or later)
R13	Quick setup	From a flashed card to the first live reading on the dashboard in 60 min or less, following the guide	Not verifiable at TRL 3	Timed trial
R14	Low cost and buildable	Parts \$200 or less; no custom PCB	Not met: \$290.00 [I1]; within the \$300 recommended budget, awaiting Amish; no custom PCB is met	Supplier quotes
R15	Open and exportable	All software open source; data exportable as CSV and through an open API	Met by design	Design review
R16	Monitoring only	No control outputs; the gateway cannot actuate the monitored system	Met by design	Design review

■ Assumptions

The assumptions behind each status are listed in TWK-CAL-001, Table 1. The main ones are:

- A sensor reading is a LoRaWAN uplink of 20 bytes of payload plus 13 bytes of overhead, at 125 kHz and coding rate 4/5 (71.9 ms at SF7, 246.8 ms at SF9, 1810.4 ms at SF12).
- Uplinks spread evenly over 8 channels and arrive at random (pure ALOHA), with no capture effect.
- A stored value takes 90 bytes including index overhead, before compression.
- The gateway draws 6.41 W on average: board 4 W, fan 0.5 W, concentrator 1 W, converter at 90 % and UPS loss 0.3 W.
- The backup pack delivers 80 % of its 19.2 Wh rating when new.
- The processor throttles at 85 °C and sits 4 K/W above the enclosure air with its fan running; both are assumptions to be measured.